

MIRAI TRACK

End Sem Examination

Problem Statement:

1. **Load & inspect** — Load `ipl_2026_deliveries.csv` into a pandas DataFrame. Print its shape, column dtypes, and check for missing values and tell if any column contains missing values. Also check statistics of the data (use `describe()`) and tell at least one insight which you can fetch from that stats table. (10 Marks)
2. **Top run-scorers** — Find the top 10 batters with the highest total `runs_of_bat` using `groupby()` on the `striker` column and visualize them with a **bar plot**. (10 Marks)
3. **Runs-per-ball distribution** — Plot a histogram of the `runs_of_bat` column. What is the most common number of runs scored off a single ball? (10 Marks)
4. **Venue-wise match count** — Count the number of unique matches (`match_id`) played per `venue` and plot all the venues using a bar chart. (10 Marks)
5. **Match-wise scoring trend** — Using `match_no`, compute the total runs scored (`runs_of_bat + extras`) per match and plot it as a **line chart** across the season. Which match had the highest run output? (15 Marks)
6. **Bowler performance segmentation** — For each `bowler`, compute: total balls bowled, total runs conceded (`runs_of_bat + extras`), total wickets taken (count of non-null `wicket_type` where the bowler is credited, i.e. excluding `runout`), and economy rate. Create a derived metric (e.g., `impact_score = wickets * log(total_balls + 1) / economy`), then use a seaborn scatterplot for (economy vs `impact_score` for top 10 `bowler` by wickets). (15 Marks)
7. **Team-wise average runs & wicket rate** — Group the data by `batting_team` using `groupby()`, compute the mean `runs_of_bat` per ball and the wicket-loss rate (wickets per ball faced), sort to find the top 5 teams by average runs per ball, now compare the average runs and wicket-loss rate in a single line graph according to `venue` (venues on x-axis, runs per ball and wicket-loss rate on y-axis) (remember runs per ball may be very high in comparison to wicket-loss rate, so you can normalize the wicket-loss rate with respect to runs per ball). (30 Marks)